MECHATRONIC SYSTEM INTEGRATION
(MCTA 3203)

SEMESTER 1, 2026/2027

EXPERIMENT 1

Digital Logic System: 7-Segement Display Arduino UNO

(MCTA 3203)

SECTION 1

GROUP 6

NO.	MATRIC NO.	NAME
1	2414523	MUHAMMAD RASYAD BIN KHAIRIL ANNUAR
2	2414509	MUHAMMAD HAZIQ HAIKAL BIN MOHD YUSRI
3	2415329	MUHAMMAD IRFAN BIN AMIR AMRAN
4	2410303	MUHAMMAD NUR IKHWAN BIN ABDUL MUKTI

SUBMISSION DATE: 12 OCTOBER 2026

Abstract

The core objective of this experiment was to help us understand how to interface a 7-segment display with an Arduino Uno and how to control it manually using buttons. In order to run individual display segments from D2 to D8 and control interactive pushbutton triggers, the experimental process involved assembling the physical circuit configuration, uploading customized control code via Serial USB communication, and running sequential software algorithms using exact 0.5 seconds timing delays. The main results achieved a consistent response with a constant half-second state transition interval and clear visual output, successfully demonstrating correct numerical counting behaviors from 0 to 5 and up to 9. There were a few challenges we had to face during the experiment. The first challenge was that the segment would not light up if the command in the code is logic HIGH and if we connect the common cathode pin of the display to the ground pin of the Arduino. Then, the reset push button could not work at first. Hence, we tried to fix our coding by changing logic HIGH to LOW and connecting the common cathode pin of the display to the 5V pin of the Arduino so the segments would light up. The coding also was edited to ensure the reset push button could work effectively according to the objective of the experiment.

Table of Contents

1.0 Introduction & Theory

1.1 Materials and Equipment

1.2 Experimental Setup

1.3 Methodology & Programming Logic

1.4 Data Collection & Analysis

1.5 Results & Discussion

1.6 Conclusion

Recommendations

References

Appendices

Acknowledgement

Student Declaration

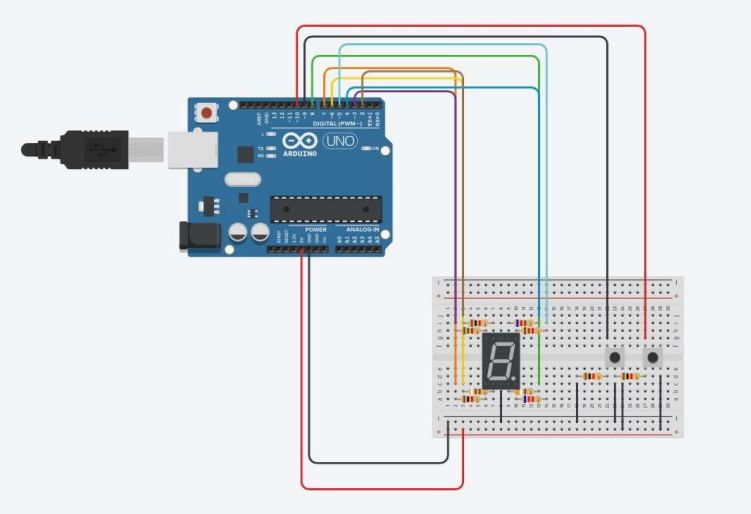
1.0 Introduction & Theory

The experiment conducted relies on a common cathode 7-segment display, which comprises seven individual light-emitting diodes (a, b, c, d, e, f, g) sharing a common ground connection linked to the Arduino ground pin. When the Arduino output pins drive the corresponding anode pins to a HIGH logic state, individual segments light up. Each segment line has 220-ohm resistors placed in series to prevent excessive current and protect the components. Additionally, pushbuttons with 10K-ohm pull-up resistors are used to deal with manual user inputs. The main objective of the experiment is to successfully connect an Arduino Uno with push buttons and a common cathode 7-segment display in order to manually control and show numerical counting sequences. A breadboard that correctly connects digital output pins (D2 to D8) to the display segments together with appropriate push button and resistor wiring is crucial to achieve the experiment's objective which intends to carry out firmware logic when the corresponding push buttons are pressed, either sequentially increases numbers or resets the count to zero. The software initializes the digital pins by setting the push button pins as inputs with suitable resistor configurations and the display segment pins as outputs.

1.1 Materials and Equipment

i. Arduino Uno board
ii. Common cathode 7-segment display
iii. 220-ohm resistors (7)
iv. 10K-ohm resistors (2)
v. Push buttons (2)
vi. Jumper wires
vii. Breadboard
viii. Arduino IDE (software)

1.2 Experimental Setup



Circuit Design Schematics using Tinkercad

1. Common cathode 7-segment display to the Arduino Uno board connections:

- Each segment pin (A–G) of the display was connected to Arduino digital pins D2–D8 through 220 Ω resistors to limit the current flow.
- The common cathode pin of the display was connected to one of the Arduino GND pins.

2. Pushbuttons to the Arduino Uno board:

- One leg of the first pushbutton was connected to Arduino pin D9 to function as the increment button.
- One leg of the second pushbutton was connected to Arduino pin D10 to act as the reset button.
- The opposite legs of both pushbuttons were connected to the ground rail of the breadboard.
- A 10 kΩ pull-up resistor was connected to the increase button, with one end attached to the corresponding digital input pin (D9) and the other end connected to the 5V line.

3. The Arduino Uno board was powered through a USB connection, providing both 5V and GND to the entire circuit.

1.3 Methodology & Programming Logic

Circuit Construction

The circuit was set up on a breadboard based on the circuit design. Each segment (A–G) of the 7-segment display is connected to the digital pins (D2-D8) of the Arduino Uno board using 220Ω resistors to limit the current. The common cathode of the display was connected to GND.

Pushbutton Connections

Two push buttons were used in this experiment. One button was used to increase the displayed number correctly and the other was used to reset the number back to zero. Each button was connected to digital pins D9 and D10 with 10kΩ pull up resistors to make the input stable. Each button was also connected to GND on one side and to 5V through the pull up resistors.

Code Upload

The Arduino IDE was used to write and upload the code to the Arduino Uno board. The program was written to control the segments so that numbers from 0 to 9 could be displayed on the 7-segment display.

When the code display number 0 to 9 sequentially:

Coding to display the number from 0 to 9 continuously while following the sequence. Delays were added to make each number stay visible for 0.5 seconds before changing.

Code upload:

```
const int segmentA = 2;
const int segmentB = 3;
const int segmentC = 4;
const int segmentD = 5;
const int segmentE = 6;
const int segmentF = 7;
const int segmentG = 8;

const int inc = 9;
const int reset = 10;
```

```
else if (number == 5)
{
  digitalWrite(segmentA, LOW);
  digitalWrite(segmentC, LOW);
  digitalWrite(segmentD, LOW);
  digitalWrite(segmentF, LOW);
  digitalWrite(segmentG, LOW);
}

else if (number == 6)
{
```

```
int count = 0;

void setup()
{
  pinMode(segmentA, OUTPUT);
  pinMode(segmentB, OUTPUT);
  pinMode(segmentC, OUTPUT);
  pinMode(segmentD, OUTPUT);
  pinMode(segmentE, OUTPUT);
  pinMode(segmentF, OUTPUT);
  pinMode(segmentG, OUTPUT);

  pinMode(inc, INPUT_PULLUP);
  pinMode(reset, INPUT_PULLUP);
}

void displayNumber(int number)
{
  // Turn everything OFF first
  digitalWrite(segmentA, HIGH);
  digitalWrite(segmentB, HIGH);
  digitalWrite(segmentC, HIGH);
  digitalWrite(segmentD, HIGH);
  digitalWrite(segmentE, HIGH);
  digitalWrite(segmentF, HIGH);
  digitalWrite(segmentG, HIGH);

  if (number == 0)
  {
    digitalWrite(segmentA, LOW);
    digitalWrite(segmentB, LOW);
    digitalWrite(segmentC, LOW);
    digitalWrite(segmentD, LOW);
    digitalWrite(segmentE, LOW);
    digitalWrite(segmentF, LOW);
  }

  else if (number == 1)
  {
    digitalWrite(segmentB, LOW);
    digitalWrite(segmentC, LOW);
  }

  else if (number == 2)
  {
    digitalWrite(segmentA, LOW);
    digitalWrite(segmentB, LOW);
    digitalWrite(segmentC, LOW);
    digitalWrite(segmentD, LOW);
    digitalWrite(segmentE, LOW);
    digitalWrite(segmentF, LOW);
    digitalWrite(segmentG, LOW);
  }

  else if (number == 7)
  {
    digitalWrite(segmentA, LOW);
    digitalWrite(segmentB, LOW);
    digitalWrite(segmentC, LOW);
  }

  else if (number == 8)
  {
    digitalWrite(segmentA, LOW);
    digitalWrite(segmentB, LOW);
    digitalWrite(segmentC, LOW);
    digitalWrite(segmentD, LOW);
    digitalWrite(segmentE, LOW);
    digitalWrite(segmentF, LOW);
    digitalWrite(segmentG, LOW);
  }

  else if (number == 9)
  {
    digitalWrite(segmentA, LOW);
    digitalWrite(segmentB, LOW);
    digitalWrite(segmentC, LOW);
    digitalWrite(segmentD, LOW);
    digitalWrite(segmentF, LOW);
    digitalWrite(segmentG, LOW);
  }
}

void loop()
{
  displayNumber(count);

  if (digitalRead(inc) == HIGH)
  {
    count++;

     if (count > 9)
     {
```

```
  digitalWrite(segmentD, LOW);
  digitalWrite(segmentE, LOW);
  digitalWrite(segmentG, LOW);
 }

 else if (number == 3)
 {
  digitalWrite(segmentA, LOW);
  digitalWrite(segmentB, LOW);
  digitalWrite(segmentC, LOW);
  digitalWrite(segmentD, LOW);
  digitalWrite(segmentG, LOW);
 }

 else if (number == 4)
 {
  digitalWrite(segmentB, LOW);
  digitalWrite(segmentC, LOW);
  digitalWrite(segmentF, LOW);
  digitalWrite(segmentG, LOW);
 }
```

```
     count = 0;
    }

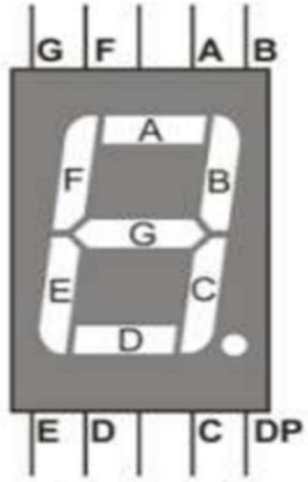
    delay(300);  // debounce
  }

  if (digitalRead(reset) == HIGH)
  {
    count = 0;

    delay(300);
  }

}
```

1.4 Data Collection & Analysis



7-Segement Display pin out

Number to display	Segments to Switch ON (HIGH)	Segments to Switch OFF (LOW)
0	a,b,c,d,e,f	g
1	b,c	a,d,e,f,g
2	a,b,d,e,g	c,f
3	a,b,c,d,g	e,f
4	b,c,f,g	a,d,e
5	a,c,d,f,g	b,e
6	a,c,d,e,f,g	b,
7	a,b,c	d,e,f,g
8	a,b,c,d,e,f,g	-
9	a,b,c,d,f,g	e

Table 1.1

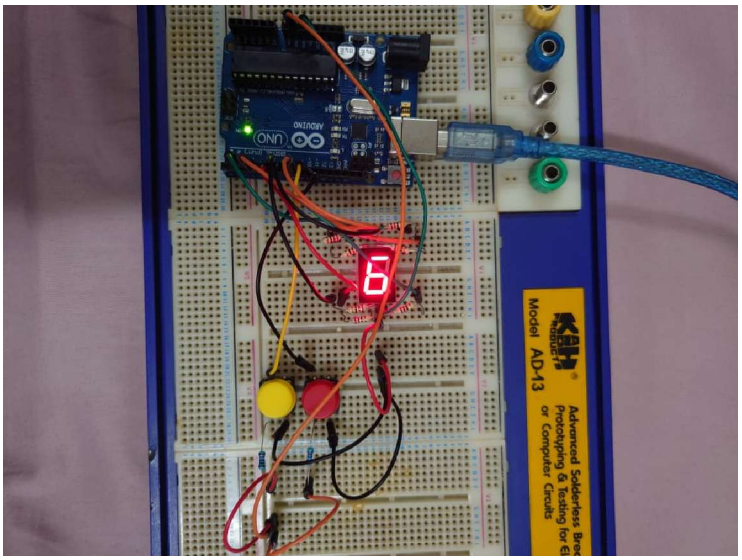
Which button pressed	Expected Display	Actual Display
-	0	0
Increment button	1	1
Increment button	2	2
Increment button	3	3
Increment button	4	4
Increment button	5	5
Increment button	6	6
Increment button	7	7
Increment button	8	8
Increment button	9	9
Reset button	0	0

Table 1.2

Based on Table 1.2, the 7-segment display successfully displayed the numbers from 0 to 9 when the increment button was pressed. Each button press increased the displayed number by one, and the counter returned 0 after reaching 9. When the reset button was pressed, the displayed number returned to 0 regardless of the previous count.

Overall, the expected outputs matched the recorded outputs, indicating that the circuit and program functioned correctly during the test.

1.5 Results & Discussion



The experimental results showed that the Arduino Uno could control the common-cathode 7-segment display through its digital output pins. Each segment was activated according to the programmed combination of HIGH and LOW signals, allowing the display to represent different numerical values.

The increment button was used to increase the counter sequentially from 0 to 9. Once the counter reached 9, the display returned 0 following the next increment. The reset button provided an additional control function by returning the counter to 0 immediately. These observations demonstrated the successful integration of the display, pushbuttons, and Arduino Uno.

One limitation observed during testing was the possibility of button bouncing. When a mechanical pushbutton is pressed, its contacts may briefly switch between ON and OFF several

times before settling. The Arduino may interpret these rapid changes as multiple button presses, causing the counter to increase unexpectedly.

Discussion: Comparison of I2C LCD, 7-Segment Display and Matrix LED Interfacing

An I2C LCD can be connected to Arduino using the SDA and SCL pins to display text and numbers. Its coding uses a library, such as LiquidCrystal_I2C, and functions like lcd.print() to display information. However, a 7-segment display requires the Arduino to control individual segments using HIGH and LOW signals to show numbers. A matrix LED controls rows and columns of LEDs to display patterns, symbols, and characters. Therefore, an I2C LCD is suitable for displaying text, while a 7-segment display is suitable for showing numbers.

1.6 Conclusion

In conclusion, this experiment successfully demonstrated the interfacing of an Arduino Uno with a common-cathode 7-segment display and pushbuttons. The counter was able to increase from 0 to 9 and return to 0 using the increment button, while the reset button successfully returned the display to 0. The experiment also improved understanding of digital input and output signals, programming logic, and hardware interfacing. However, button bouncing may affect the accuracy of the counter, which can be improved by implementing a debouncing technique. Overall, the experiment achieved its objectives and highlighted the differences between I2C LCD, 7-segment display, and matrix LED interfacing, as well as their applications in displaying information.

Recommendations

- Improve Circuit Connections:

  Ensure that all components are properly connected and secured to prevent loose connections that may affect the performance of the 7-segment display.

- Expand the Counting Range:

  An additional 7-segment display can be added to allow the counter to display numbers beyond 9, such as 00 to 99.

- Improve Pin Efficiency:

  A display driver IC, such as the MAX7219, can be used to reduce the number of Arduino digital pins required to control the display.

References

- Autodesk. (n.d.). *Tinkercad circuits*. https://www.tinkercad.com/circuits
- Arduino. (n.d.). *Arduino IDE*. https://www.arduino.cc/en/software

Appendices

A- Arduino Codes

MSI_lab1_numbercounter | Arduino IDE 2.3.10

File Edit Sketch Tools Help

Arduino UNO

MSI_lab1_numbercounter.ino

```
const int segmentA = 2;
const int segmentB = 3;
const int segmentC = 4;
const int segmentD = 5;
const int segmentE = 6;
const int segmentF = 7;
const int segmentG = 8;

const int inc = 9;
const int reset = 10;

int count = 0;

void setup()
{
  pinMode(segmentA, OUTPUT);
  pinMode(segmentB, OUTPUT);
  pinMode(segmentC, OUTPUT);
  pinMode(segmentD, OUTPUT);
  pinMode(segmentE, OUTPUT);
  pinMode(segmentF, OUTPUT);
  pinMode(segmentG, OUTPUT);

  pinMode(inc, INPUT_PULLUP);
  pinMode(reset, INPUT_PULLUP);
}

void displayNumber(int number)
{
  // Turn everything OFF first
  digitalWrite(segmentA, HIGH);
  digitalWrite(segmentB, HIGH);
  digitalWrite(segmentC, HIGH);
  digitalWrite(segmentD, HIGH);
  digitalWrite(segmentE, HIGH);
  digitalWrite(segmentF, HIGH);
  digitalWrite(segmentG, HIGH);

  if (number == 0)
  {
    digitalWrite(segmentA, LOW);
    digitalWrite(segmentB, LOW);
    digitalWrite(segmentC, LOW);
    digitalWrite(segmentD, LOW);
    digitalWrite(segmentE, LOW);
    digitalWrite(segmentF, LOW);
  }

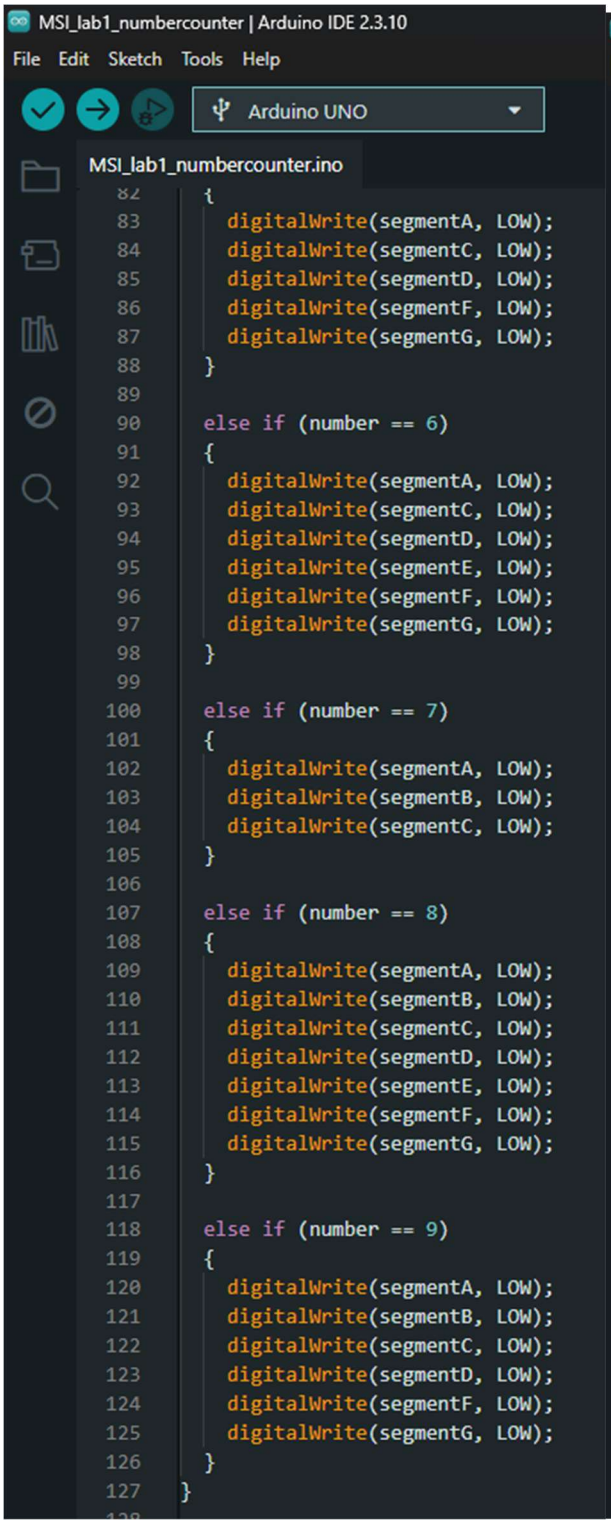
  else if (number == 1)
  {
    digitalWrite(segmentB, LOW);
    digitalWrite(segmentC, LOW);
  }

  else if (number == 2)
  {
    digitalWrite(segmentA, LOW);
    digitalWrite(segmentB, LOW);
    digitalWrite(segmentD, LOW);
    digitalWrite(segmentE, LOW);
    digitalWrite(segmentG, LOW);
  }

  else if (number == 3)
  {
    digitalWrite(segmentA, LOW);
    digitalWrite(segmentB, LOW);
    digitalWrite(segmentC, LOW);
    digitalWrite(segmentD, LOW);
    digitalWrite(segmentG, LOW);
  }

  else if (number == 4)
  {
    digitalWrite(segmentB, LOW);
    digitalWrite(segmentC, LOW);
    digitalWrite(segmentF, LOW);
    digitalWrite(segmentG, LOW);
  }

  else if (number == 5)
  {
    digitalWrite(segmentA, LOW);
    digitalWrite(segmentC, LOW);
    digitalWrite(segmentD, LOW);
    digitalWrite(segmentF, LOW);
    digitalWrite(segmentG, LOW);
  }
```



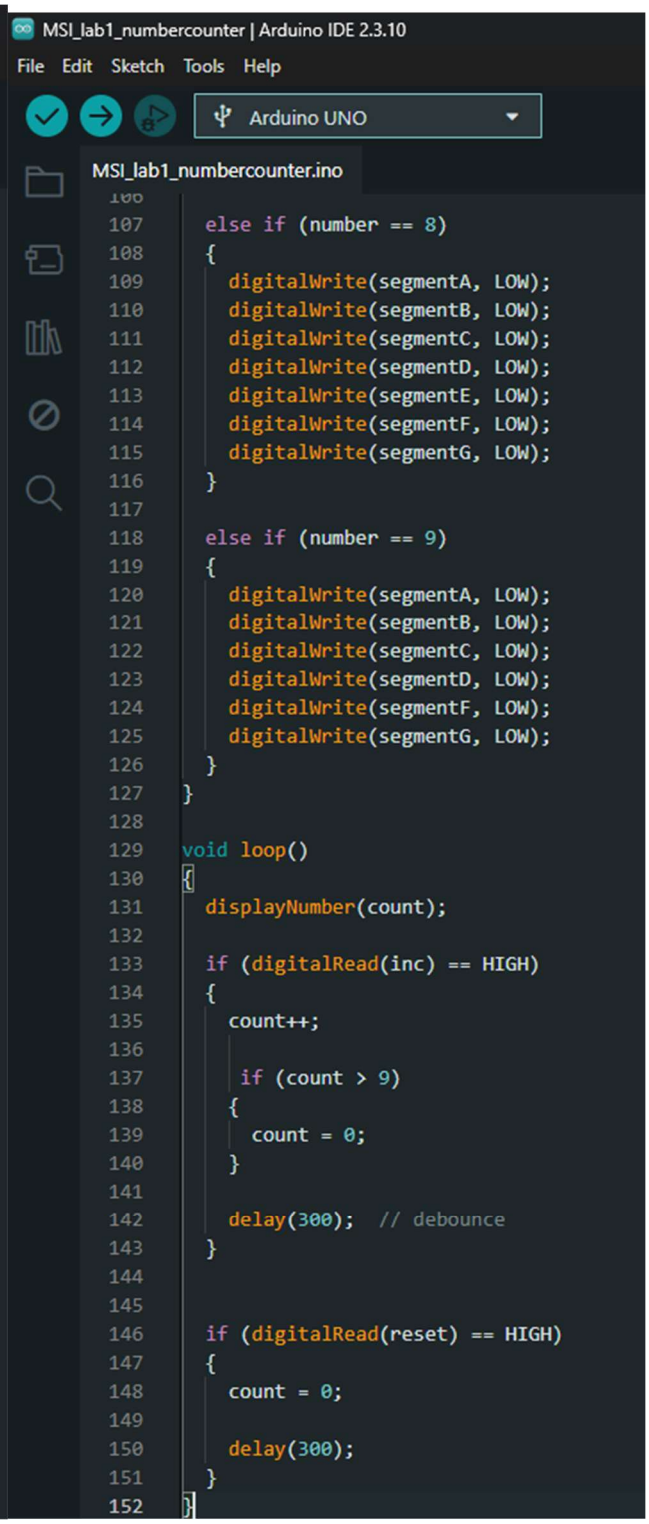

```
    {
      digitalWrite(segmentA, LOW);
      digitalWrite(segmentC, LOW);
      digitalWrite(segmentD, LOW);
      digitalWrite(segmentF, LOW);
      digitalWrite(segmentG, LOW);
    }

    else if (number == 6)
    {
      digitalWrite(segmentA, LOW);
      digitalWrite(segmentC, LOW);
      digitalWrite(segmentD, LOW);
      digitalWrite(segmentE, LOW);
      digitalWrite(segmentF, LOW);
      digitalWrite(segmentG, LOW);
    }

    else if (number == 7)
    {
      digitalWrite(segmentA, LOW);
      digitalWrite(segmentB, LOW);
      digitalWrite(segmentC, LOW);
    }

    else if (number == 8)
    {
      digitalWrite(segmentA, LOW);
      digitalWrite(segmentB, LOW);
      digitalWrite(segmentC, LOW);
      digitalWrite(segmentD, LOW);
      digitalWrite(segmentE, LOW);
      digitalWrite(segmentF, LOW);
      digitalWrite(segmentG, LOW);
    }

    else if (number == 9)
    {
      digitalWrite(segmentA, LOW);
      digitalWrite(segmentB, LOW);
      digitalWrite(segmentC, LOW);
      digitalWrite(segmentD, LOW);
      digitalWrite(segmentF, LOW);
      digitalWrite(segmentG, LOW);
    }
}
```




```
    else if (number == 8)
    {
      digitalWrite(segmentA, LOW);
      digitalWrite(segmentB, LOW);
      digitalWrite(segmentC, LOW);
      digitalWrite(segmentD, LOW);
      digitalWrite(segmentE, LOW);
      digitalWrite(segmentF, LOW);
      digitalWrite(segmentG, LOW);
    }

    else if (number == 9)
    {
      digitalWrite(segmentA, LOW);
      digitalWrite(segmentB, LOW);
      digitalWrite(segmentC, LOW);
      digitalWrite(segmentD, LOW);
      digitalWrite(segmentF, LOW);
      digitalWrite(segmentG, LOW);
    }
}

void loop()
{
  displayNumber(count);

  if (digitalRead(inc) == HIGH)
  {
    count++;

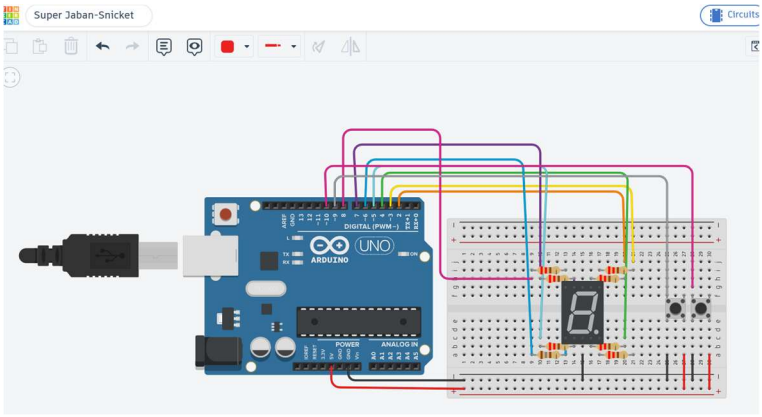
     if (count > 9)
    {
      count = 0;
    }

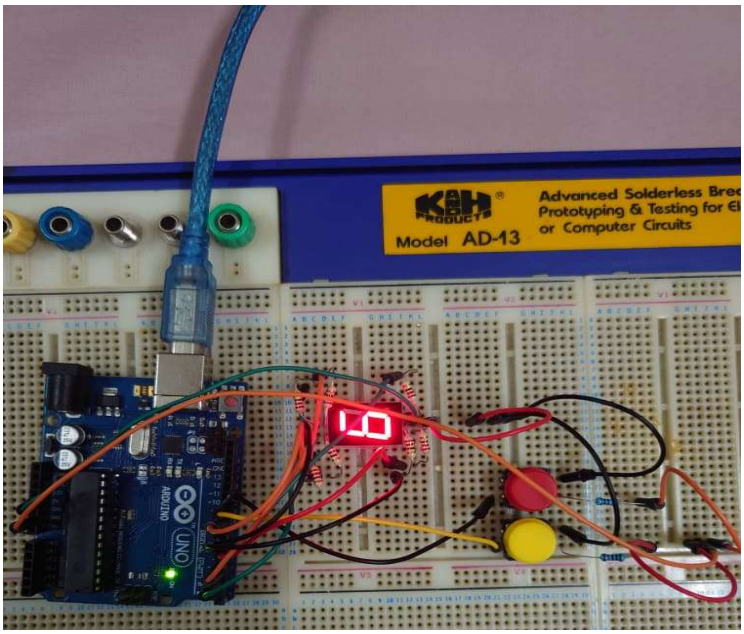
    delay(300);  // debounce
  }


  if (digitalRead(reset) == HIGH)
  {
    count = 0;

    delay(300);
  }
}
```

B- Circuit Diagram





Acknowledgement

We would like to express our deepest appreciation to our instructor and laboratory personnel for their guidance, support, and assistance throughout this experiment. Their explanations and feedback helped us understand the practical application of microcontroller programming, sensor communication, and data acquisition. We would also like to thank our group members for their cooperation and contribution during the preparation, assembly, testing, and analysis of the experimental system.

Student's Declaration

We hereby declare that the work presented in this technical report is entirely our own, except where explicitly acknowledged. We certify that in developing and completing this mechatronics project, we have strictly adhered to the standards of academic integrity and have not engaged in any form of plagiarism or unethical conduct. All sources of information, reference materials, and technical support utilized in this work have been appropriately cited and acknowledged.

Certificate of Originality and Authenticity

This is to certify that we are responsible for the work submitted in this report, that the original work is our own except as specified in the references and acknowledgment, and that the original work contained herein has not been undertaken or done by unspecified sources or persons. We hereby certify that this report has not been done by only one individual, and all of us have contributed to the report. The length of contribution to the reports by each individual is noted within this certificate. We also hereby certify that we have read and understand the content of the total report, and no further improvement on the reports is needed from any of the individual contributors to the report. We therefore agreed unanimously that this report shall be submitted for marking, and this final printed report has been verified by us.

Name: MUHAMMAD RASYAD BIN KHAIRIL ANNUAR Agree **[/]**
Matric: 2414523

Name: MUHAMMAD HAZIQ HAIKAL BIN MOHD YUSRI Agree **[/]**
Matric: 2414509

Name: MUHAMMAD IRFAN BIN AMIR AMRAN Agree **[/]**
Matric: 2415329

Name: MUHAMMAD NUR IKHWAN BIN ABDUL MUKTI Agree **[/]**
Matric: 2410303